

MASFE: An Onboard Multi-Algorithm Scheduling and Fusion Engine

For Regenerative Agriculture SmallSat Missions

Emil Wes Lambert, BSc Aerospace Engineering, TU Delft
Solo submission

Abstract. MASFE moves regenerative-agriculture monitoring onboard by screening every pass with lightweight EVI/NDWI cues, escalating likely stress through CSC fusion, and downlinking only alert tiles. In a submitted 100-seed Monte Carlo benchmark, MASFE achieves 99.3% downlink reduction versus raw, 38.6% energy savings, 100% disease-event recall, and a 1.6% false-positive rate within the selected payload budget. A Westlands 2024 MODIS replay supports the scheduler on real scenes. The remaining gap is hardware-side validation.

1 Introduction

MASFE (Multi-Algorithm Scheduling and Fusion Engine) answers NASA’s Space-to-Soil challenge by moving land-observation orchestration onboard for regenerative agriculture, where crop stress must be detected early enough for patch-level intervention instead of blanket chemical application [1, 2, 3, 4]. Current land products are often too slow or coarse for that workflow.

The proposed contribution is a two-stage, MDP-formulated scheduling policy deployed as a rule-based controller on a LEON4FT processor; Fig. 1 and §2 summarize the architecture and implementation.

Table 1: MASFE response to all five judging criteria.

Criterion	Where	Key evidence
Creativity	§2–3	Beta-posterior scheduler advancing EO-1 ASE heritage [5]; complementary cadence to Dynamic Targeting [6].
Feasibility	§2–3	SWAP closure, compute margin, and real-scene replay.
Impact	§3	Monte Carlo reduction, recall, and false-positive evidence.
Business	§4	5-year rollout and multi-platform distribution.
Presentation	Eq. (1), Fig. 1, code	Architecture, CSC equation, reproducible simulation submission.

2 Proposed Solution Detail

MASFE uses a two-stage adaptive sensing loop. Stage 1 executes an EVI anomaly screen derived from MOD13A1 logic on every observed field patch and updates a per-patch stress posterior using EVI and NDWI cues from the same MSI acquisition. When the posterior indicates likely stress, Stage 2 runs a thermal retrieval derived from MOD11A1 and fuses EVI, LST, and MOD09A1-derived NDWI into a Crop Stress Composite (CSC), a normalized anomaly index conceptually related to temperature–vegetation dryness indices (e.g., TVDI) [3, 4, 7, 8]. Only confirmed anomalous tiles are buffered for priority downlink. This makes sensing content-driven rather than timeline-driven and distinguishes MASFE from ASPEN/CASPER-style planners, which optimize scheduled activities rather than per-patch stress belief [9, 10]. EO-1 ASE demonstrated rule-based change-detection triggers in orbit [5]; MASFE advances that heritage with per-patch Beta posteriors that accumulate evidence across passes, enabling stress-trajectory-sensitive scheduling rather than single-pass exceedance detection. Candela et al.’s Dynamic Targeting instead focuses on rapid retargeting of pointable instruments toward transient high-value scenes; MASFE is complementary in operating at composite cadence for chronic crop stress that must be accumulated across passes [6].

$$\text{CSC} = 0.40 \cdot \text{clip}\left(\frac{\Delta\text{EVI}}{5\sigma_e}, 0, 1\right) + 0.35 \cdot \text{clip}\left(\frac{\Delta\text{LST}}{4\sigma_l}, 0, 1\right) + 0.25 \cdot \text{clip}\left(\frac{\Delta\text{NDWI}}{4\sigma_n}, 0, 1\right) \quad (1)$$

where $\Delta\text{EVI} = \max(0, \text{EVI}_{\text{base}} - \text{EVI}_{\text{obs}})$, $\Delta\text{LST} = \max(0, \text{LST}_{\text{obs}} - \text{LST}_{\text{base}})$, and

$\Delta\text{NDWI} = \max(0, \text{NDWI}_{\text{base}} - \text{NDWI}_{\text{obs}})$. The factors of 5, 4, and 4 are simulation-calibrated saturation constants that map anomaly magnitude into $[0, 1]$ before CSC composition; field recalibration against crop-specific agronomy is a pre-TRL-4 item.

The scheduling logic is modeled as a finite-horizon MDP with per-patch Beta posteriors $\text{Beta}(\alpha_i, \beta_i)$ over stress probability. Stage 1 increments α_i when EVI or NDWI exceedance is observed and β_i otherwise, with the evidence window capped to preserve responsiveness. Action selection compresses that patchwise belief into a mission state vector $s = (\mu_{\text{max}}, \tau_{\text{max}}, \text{SOC}, d, k)$, where μ_{max} is the maximum posterior mean, τ_{max} is the maximum $P(p > 0.5)$ tail probability, SOC is battery state of charge, $d \in \{0, 1\}$ indicates downlink availability, and k counts passes since last fusion. The action set remains $\{\text{SKIP}, \text{MOD13}, \text{FUSE}, \text{FUSE_PRIORITY}\}$; NDWI is derived from the same MSI acquisition rather than requiring a separate action. The reward trades information gain against energy cost and missed-alert risk. The deployed solver is a hand-tuned deterministic policy chosen for flight-V&V traceability.

Table 2: Mission requirements and demonstrated MASFE results.

Requirement	MASFE response
Alert latency (observation pass to first ground contact) ≤ 90 min	One-pass onboard screening and fusion followed by X-band priority downlink at the first available contact.
Spatial resolution ≤ 30 m GSD	Tiered GSD pyramid: 30 m screening on all passes, 10 m confirmation on suspect patches, and 4.6 m native evidence tiles for priority alerts.
False-positive rate $\leq 15\%$	1.6% with benign confounders and roughly 30% cloud-obscured passes per seed (§3).
SmallSat SWAP compliance	9.6 W peak / 5.05 W duty-cycled payload average; 92.5% peak LEON4FT ceiling with 39.2% Stage-1 and 49.2% seasonal-average compute; 86% observed benchmark minimum SOC in a local 40 Wh battery buffer.
Downlink efficiency	99.3% reduction versus raw downlink baseline while retaining 100% disease-event recall (§3).

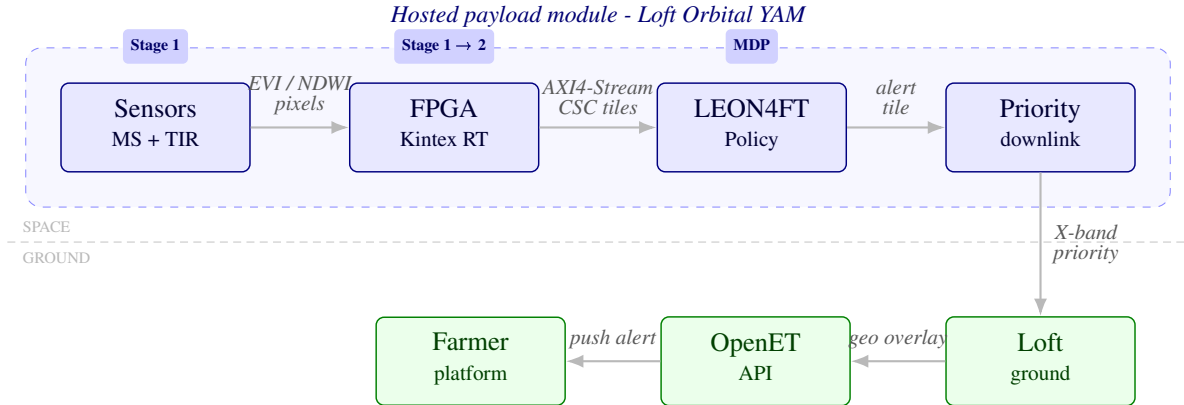


Figure 1: MASFE end-to-end architecture. The onboard payload (dashed box) runs on a hosted module aboard Loft Orbital’s YAM bus. Stage 1 EVI/NDWI screening executes every pass; Stage 2 thermal fusion fires only on stress-confirmed patches; priority alert tiles are downlinked X-band and delivered to farm software via the OpenET API.

The FPGA performs radiometric preprocessing and per-pixel CSC computation, then transfers frames to the LEON4FT over AXI4-Stream (budgeted under 1 ms). Onboard preprocessing includes radiometric calibration and cloud screening, while CSC itself is evaluated as a pass-to-pass relative anomaly detector rather than a fully atmosphere-corrected retrieval; full atmospheric correction is deferred to later validation because CSC is used for change detection rather than absolute retrieval. The selected Kintex UltraScale+

RT part comes from AMD’s radiation-tolerant XQR family [11]. MASFE is framed as a hosted payload module within a Loft Orbital YAM bus rather than as a full standalone 6U satellite; the SWAP budget therefore describes the payload module itself, not the host spacecraft.

Operational degradation is explicit. Between 15% and 35% battery state of charge, the degraded mode preserves Stage 1-only screening; below 15%, it preserves alert storage and downlink. This satisfies the challenge requirement to degrade gracefully under onboard resource stress [1].

Processor utilization remains below the worst-case ceiling and preserves margin for RTOS and fault-handling overhead.

Each alert packet contains a geo-tagged field-patch polygon with centroid coordinates, CSC severity score, contributing EVI/LST/NDWI anomaly components, timestamp and pass ID, and a confidence flag derived from consecutive confirming passes. Escalated alert tiles are compressed onboard using CCSDS 122.0 wavelet coding with a planning assumption of roughly 2.5:1, reducing a nominal 8.4 MB/km² native alert stream to about 3.36 MB/km² delivered before packetization [12]. Per-patch geolocation is derived from bus ephemeris and attitude solutions and is budgeted to remain comfortably inside the 30 m effective-GSD requirement. These packets are delivered through Loft Orbital’s ground network to farm software as geo-tagged overlays via the OpenET-linked API, enabling push notifications within 15 minutes after first ground contact.

Full simulation source code is available as Code Component 3 at github.com/emillambert/Emil_MASFE.

3 Potential Impact

Dr. Katie Gold’s disease-detection work emphasizes that spectral stress often appears before visible symptoms and before whole-field treatment would be agronomically or economically rational [2].

In a 100-seed Monte Carlo benchmark, MASFE was tested over 120 passes, 500 field patches, 21 disease events, and 9 benign confounders per seed, with roughly 36.7 cloud-obscured passes per seed.

Across that benchmark, 95% confidence intervals were 99.32–99.33% for downlink reduction versus raw, 38.52–38.70% for energy saving versus raw, 100.0–100.0% for disease-event recall, and 1.34–1.78% for false-positive rate. The downlink-reduction interval is exceptionally tight because the raw baseline is fixed and MASFE’s total byte count is dominated by stable Stage-1 screening traffic, so seed-to-seed variability mainly changes alert counts rather than total data volume. The operating point is tunable: sweeping *csc_alert_thr* from 0.40 to 0.70 preserved 100.0% recall until the most conservative endpoint while moving false-positive rate from 2.23% to 1.26%, and a $\pm 30\%$ CSC-parameter sweep kept recall at 100.0% with false-positive rate in the 1.0–2.3% band; detailed curves and cases are included in the code submission.

Table 3: Baseline comparisons under the synthetic Monte Carlo benchmark.

Variant	Downlink ↓	Recall ↑	FP ↓	Compute ↓
Raw downlink	0.0%	100.0%	1.6%	N/A
Fixed-period onboard	96.5%	100.0%	1.6%	92.5%
EVI-only threshold	98.1%	100.0%	2.3%	39.2%
No-belief MASFE	96.7%	100.0%	1.7%	76.2%
Full MASFE	99.3%	100.0%	1.6%	49.2%

Raw downlink denotes no onboard triage; recall and false-positive rate apply the same detector after ground-processing the full dataset.

Belief memory drives the 27 pp compute reduction; NDWI fusion drives FP from 2.3% to 1.6%; staged confirmation preserves 100% recall; onboard filtering drives the 99.3% downlink reduction.

The code component also replays official MOD13A1 and MOD11A1 subsets over Westlands/Firebaugh, California, from June through October 2024, aligning MOD13A1 composite dates to a 1 km grid,

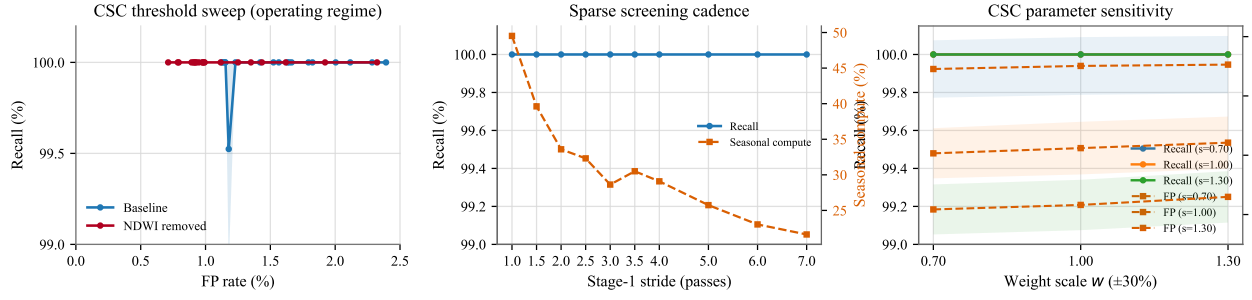


Figure 2: Additional ablations and curves (95% CI bands where applicable). Left: operating-point sweep comparing baseline CSC to an NDWI-removed fallback over the relevant FP band; more permissive thresholds eventually drive recall collapse through battery-driven CONSERVE mode. Middle: sparse Stage 1 cadence, where recall stays near ceiling while *seasonal-average compute* (right axis) shows the cost of under-sampling. Right: $\pm 30\%$ CSC weight/saturation grid (w =weight scale, s =saturation scale); recall stays within 99–100% while *FP rate* (right axis) varies across calibration corners.

collapsing MOD11A1 to median clear-sky daytime LST over the same 16-day window, and applying the unchanged MASFE policy as a real-scene scheduler validation rather than a labeled disease benchmark. The replay path accepts MOD09A1-derived NDWI where available, but the tracked Westlands anchor bundle predates that extension and therefore falls back cleanly to EVI/LST fusion on the cached artifacts. Across 7 valid composite windows after a 3-window warmup, MASFE issued one confirmed FUSE_PRIORITY alert on July 27, 2024 and six additional FUSE-only windows at 99.5% mean valid coverage. These results are software-demonstrated today and require hardware-side validation before TRL 4.

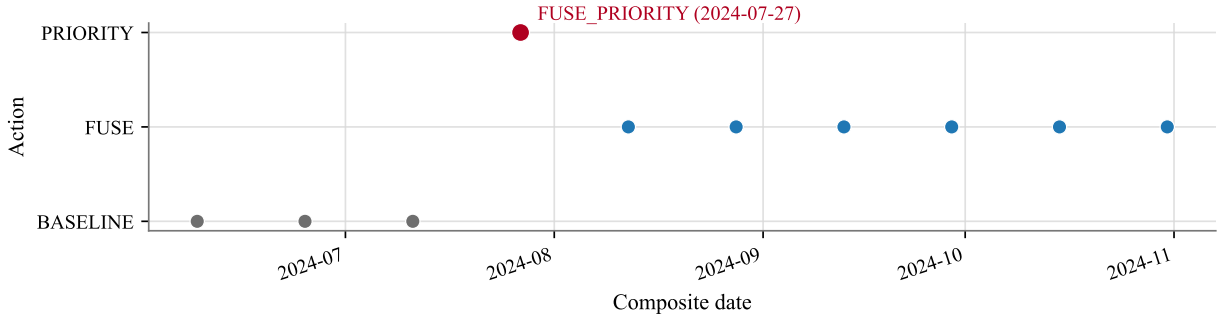


Figure 3: Westlands 2024 replay action timeline showing baseline windows, FUSE windows, and the single FUSE_PRIORITY alert on July 27, 2024.

The farm-scale value proposition is concrete. For a 500 ha regenerative tomato operation, UC Cooperative Extension processing-tomato budgets place representative disease-control line items in the rough \$74–119/ha/season range, implying roughly \$34,000–\$55,000/season of avoidable chemical spend if MASFE confines treatment to the roughly 8% of the field flagged during an event [13]. That matters because late blight remains a destructive tomato disease that can destroy entire fields and collapse plants within 7–10 days under cool, wet conditions [14]. Against a global fungicide-and-bactericide baseline of roughly 0.8 Mt active ingredient per year in 2023, MASFE’s patch-level triage model changes how fungicide demand is generated, not just farm-level cost [15].

4 Team Experience and Market Evaluation

Emil Wes Lambert is completing a BSc in Aerospace Engineering at TU Delft, with coursework in spacecraft systems, AOCS, mission analysis, and embedded flight software. At AeroDelft he has led

partnerships and business operations, including procurement, sponsor relations, and participation in the EUR 73 M HAPSS hydrogen-propulsion consortium alongside Airbus, KLM, and NLR. That background supports MASFE's SWAP realism, AIST-track funding strategy, and commercial positioning.

California is the validation anchor and initial go-to-market wedge. The same architecture scales from SJV into the wider U.S. irrigated base, then into EU and Brazilian irrigation markets, and finally into a global irrigated-farmland platform with the same onboard scheduler and region-specific recalibration of CSC thresholds [16, 17, 18, 19]. The paper-facing economics therefore use a low/design-to-cost case, a \$4/ha/year alerting product, a modeled 90% annual retention rate, and a Year 2 soil-carbon MRV add-on aligned to Verra VM0042-style workflows [20, 21].

Table 4: Illustrative MASFE rollout from SJV pilot to global platform.

Year	Milestone	Geography	Coverage	Revenue	Op. margin
Y1	SJV pilot	California SJV	3.5% SJV	\$0.20 M	grant-funded
Y2	California scale	California	28.3% SJV	\$1.70 M	17.1%
Y3	U.S. national	U.S. irrigated	11.0% US	\$10.62 M	60.4%
Y4	EU + Brazil entry	EU + Brazil	35.1% EU+Brazil	\$27.62 M	68.1%
Y5	Global platform	Global	6.0% global	\$76.50 M	70.7%

In this rollout, Y1 is a paid pilot; Y2 scales through Climate FieldView API integration plus direct enterprise sales; Y3 targets U.S.-scale profitability via Climate FieldView and John Deere Operations Center; Y4 expands through xarvio and John Deere Brazil; and Y5 targets global multi-platform distribution that could also include CropX. These platforms are API integration targets rather than committed channels, and no agreements are yet in place. Y1 is intentionally modeled as an AIST- or NASA SBIR/STTR-style co-funded bridge-to-MVP year rather than a self-financing steady state [22, 23]. In the retained fixed-cost sensitivity, the low case breaks even at Y2 and 400 kha, while the base and high cases still break even at Y3 and 2.5 Mha.

Agriculture is the entry market, but the scheduler can also be retuned for wildfire fuel-moisture monitoring in a broader fire-detection market projected to reach about \$55 B by 2030 [24]; flood-front tracking would swap CSC for water-extent products, and urban heat-island response would replace farm polygons with city blocks while reusing the anomaly-driven scheduler.

5 Expected Next Steps

MASFE is currently a TRL 3 concept: the scheduling logic, fusion equation, SWAP closure, and real-scene replay are demonstrated in software, but the payload has not yet been exercised on target hardware. The next milestone is a TRL 4 hardware-in-the-loop demonstration on a LEON4FT-class platform under NASA AIST using the replayed EarthData scene set. That work should also add FPGA-side Fmask-style Stage 0 cloud screening. After that, the clearest TRL 5 path is a hosted-payload pilot through Loft Orbital, with external validation focused on Y1 pilot assumptions, agronomy review, and channel plausibility.

If selected as a finalist, the immediate pre-pitch actions are a live MODIS replay demo, a first-pass Loft Orbital cost estimate, and two farmer interviews on latency and pricing assumptions.

6 Conclusion

MASFE is a credible Space-to-Soil solution: in the current benchmark it achieves 99.3% downlink reduction versus raw, 38.6% energy savings, 100% disease-event recall, and a 1.6% false-positive rate while meeting stated latency and payload-feasibility constraints. Released under the MIT License and built on free NASA MODIS products, it is deployable by public food-security programmes without new sensing infrastructure; Mutanga et al. document 30–60% maize yield losses from grey leaf spot in sub-Saharan Africa and note that fungicides are often sprayed uniformly because timely patch-level disease maps are unavailable [25]. The remaining gap is hardware validation, not conceptual viability.

References

- [1] NASA Earth Science Technology Office. Space-to-soil challenge: Challenge brief and guidelines. <https://nasa-space-to-soil.org>, 2026. Accessed April 2026.
- [2] Katie M. Gold. Spectral early detection of crop pathogens for regenerative agriculture applications. NASA Space-to-Soil Challenge Informational Webinar 1, 2026. Cornell University Plant Disease Diagnostics Laboratory, February 12, 2026.
- [3] NASA EOSDIS Land Processes DAAC. MOD13A1 MODIS/Terra Vegetation Indices 16-Day L3 Global 500 m SIN Grid V061. <https://doi.org/10.5067/MODIS/MOD13A1.061>, 2021.
- [4] NASA EOSDIS Land Processes DAAC. MOD11A1 MODIS/Terra Land Surface Temperature/Emissivity Daily L3 Global 1 km SIN Grid V061. <https://doi.org/10.5067/MODIS/MOD11A1.061>, 2021.
- [5] Rob Sherwood, Steve Chien, et al. Intelligent systems in space: The eo-1 autonomous sciencecraft. <https://ntrs.nasa.gov/citations/20060044325>, 2005.
- [6] Alberto Candela, Jason Swope, and Steve Chien. Dynamic targeting to improve earth science missions. *Journal of Aerospace Information Systems*, 20(11):679–689, 2023. doi:10.2514/1.I011233.
- [7] E. Vermote. MODIS/Terra Surface Reflectance 8-Day L3 Global 500m SIN Grid V061. <https://doi.org/10.5067/MODIS/MOD09A1.061>, 2021. NASA Land Processes Distributed Active Archive Center.
- [8] MODIS Land Surface Reflectance Science Computing Facility. MODIS Surface Reflectance User’s Guide, Collection 6. https://www.earthdata.nasa.gov/s3fs-public/2025-04/MOD09_User_Guide_V61.pdf, 2020.
- [9] Steve Chien et al. Aspen—automated planning and scheduling for space mission operation. <https://ntrs.nasa.gov/citations/20060040446>, 2000. Conference paper, NASA NTRS.
- [10] Daniel Tran et al. Flight software issues in onboard automated planning: Lessons learned on eo-1. <https://ntrs.nasa.gov/citations/20060043041>, 2004. International Workshop on Planning and Scheduling for Space, NASA NTRS.
- [11] AMD. Kintex ultrascale xqr radiation-tolerant fpga family. <https://www.amd.com/en/products/adaptive-socs-and-fpgas/fpga/kintex-ultrascale-xqr.html>, 2026.
- [12] European Space Agency. Ccsds image data compression asic. https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Data_Processing/CCSDS_Image_Data_Compression_ASIC, 2026.
- [13] University of California Agriculture and Natural Resources. Sample costs to produce processing tomatoes, sub-surface, drip irrigated (sdi) in the sacramento valley and northern delta, 2023. https://coststudyfiles.ucdavis.edu/uploads/pub/2023/08/21/amended_82123_2023processtomatossacramentovalleysdifinal.pdf, 2023.
- [14] University of Wisconsin Horticulture. Late blight. <https://hort.extension.wisc.edu/articles/late-blight/>, 2024.
- [15] Food and Agriculture Organization of the United Nations. Pesticides use and trade. 1990–2023. <https://www.fao.org/faostat/en/#data/RP>, 2025.
- [16] USDA National Agricultural Statistics Service. 2022 census of agriculture: United states summary and state data. https://www.nass.usda.gov/Publications/AgCensus/2022/Full_Report/Volume_1%2C_Chapter_1_US/usv1.pdf, 2024.
- [17] Eurostat. Farms and farmland in the european union. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Farms_and_farmland_in_the_European_Union, 2025. Statistics Explained.
- [18] Agencia Nacional de Aguas e Saneamento Basico. Atlas irrigacao. <https://www.ana.gov.br/atlasirrigacao/>, 2021.
- [19] FAO AQUASTAT. Irrigated areas and water use: Conclusions. <https://www.fao.org/aquastat/en/data-analysis/irrig-water-use/conclusions/>, 2025.
- [20] Climate FieldView. Fieldview pricing. <https://climate.com/pricing>, 2026.
- [21] Verra. Vm0042 methodology for improved agricultural land management. <https://verra.org/methodologies/vm0042-methodology-for-improved-agricultural-land-management-v2-0/>, 2025.
- [22] NASA Earth Science Technology Office. Advanced information systems technology and esto funding process. <https://esto.nasa.gov/funding-process/>, 2026.
- [23] NASA. Small business innovation research / small business technology transfer. https://www.nasa.gov/sbir_sttr/, 2026.
- [24] Grand View Research. Global fire detection market size & outlook, 2030. <https://www.grandviewresearch.com/horizon/outlook/fire-detection-market-size/global>, 2026.
- [25] Onesimo Mutanga, Timothy Dube, and Omer Galal. Remote sensing of crop health for food security in africa: Potentials and constraints. *Remote Sensing Applications: Society and Environment*, 8:231–239, 2017. doi:10.1016/j.rsase.2017.10.004.